

Dylan Hammond

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EDUCATION

M.S. Mechanical Engineering: Utah State University (Guidance, Navigation, & Control Emphasis) **Graduated May 2026**
B.S. Mechanical Engineering: Utah State University **Graduated May 2025**

- Coursework:** Multivariable Control, Nonlinear Control, Optimal Spacecraft Guidance, Mechatronics, Robotics Planning **GPA: 3.77**

SKILLS

- Programming:** Python, MATLAB, Simulink, C++, ROS2, Git
- GNC & Robotics:** PID, LQR, Kalman Filter, Embedded Systems
- Tools & Software:** Abaqus, SolidWorks
- Hardware & Manufacturing:** 3D Printing, Carbon Fiber Manufacturing, GD&T, Technical Drawings

PROJECTS

EKF Localization Robot **May 2026 – July 2026**

- Built a ROS2 differential drive robot simulation, fusing GPS, IMU, and wheel encoder measurements through a hand-derived Extended Kalman Filter for real-time pose estimation
- Implemented the full sensor simulation, EKF, and closed-loop controller pipeline from scratch, validated in RViz

Autonomous Target-Tracking Turret **May 2025 – October 2025**

- Designed autonomous pan-tilt turret using computer vision and servo actuation for real-time target tracking
- Implemented closed-loop control system using PID controllers for stable tracking performance
- Integrated OpenCV object detection with embedded motor control on Arduino

Capstone Project: Hand-Signal Interpreter Glove **August 2024 – April 2025**

- Collaborated with team of five to develop a wearable glove capable of interpreting 25 military hand signals at 95% accuracy
- Programmed and calibrated IMU and flex sensors for real-time gesture data acquisition
- Designed and deployed a 1D CNN, enabling embedded gesture classification on an ESP32 microcontroller
- Developed embedded firmware to wirelessly control an RC vehicle based on recognized hand signals

Aggie Maps **November 2025 – December 2025**

- Developed a Python-based interactive path planning tool for pedestrian navigation on Utah State University campus
- Implemented A* algorithm to compute optimal routes with dynamical cost functions for environmental constraints

WORK EXPERIENCE

Mechanical Engineer **August 2026 – Present**
Hill Air Force Base Clearfield, UT

- Design and machine SolidWorks tooling for hydraulic actuator maintenance, including a custom holding fixture that speeds up technician teardown, repair, and test

Undergraduate Research Assistant **August 2024 – August 2025**
EMAC Lab - Utah State University Logan, UT

- Developed FE models in Abaqus to simulate electromechanical loading on materials for roadway electrification
- Automated simulation post-processing using Python, reducing analysis time

Product Development Intern **May 2025 – August 2025**
Autoliv Ogden, UT

- Optimized passenger airbag tether sizes to improve deployment safety through design validation and deployment testing
- Replicated passenger airbag tear failure modes through physical testing and analysis to support root-cause investigation

Photovoltaics and Materials Intern **May 2022 – August 2022**
Sandia National Laboratories Albuquerque, NM

- Optimized microgrid solar array power output through orientation positioning and axis tracking in Python
- Authored Sandia Labs technical paper on solar microgrid implementation strategies for small communities